

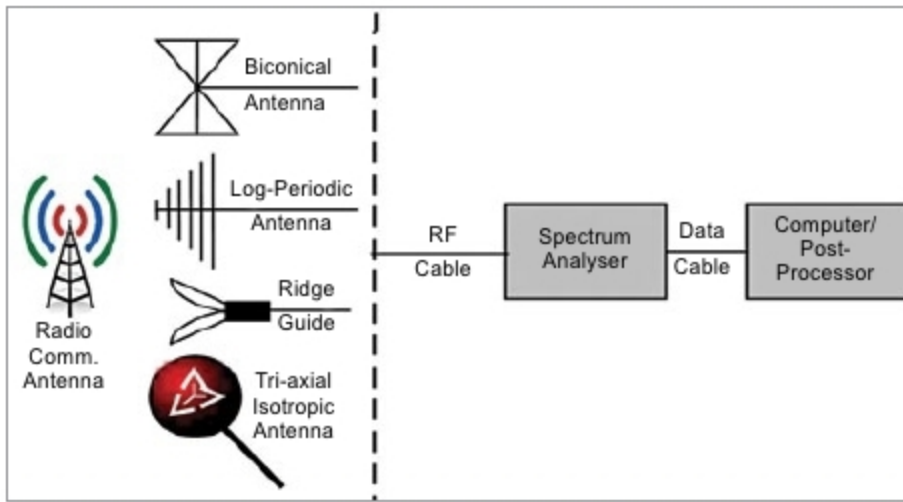


Fig. 5: Electric (E) field measurement setup

selective intelligent instruments are required. To measure the effect of any particular radiation source, all channels of the source should be transmitting at full power. For example, in the case of GSM, measurement is to be done when all voice channels are running at full load (all control channels are continuously radiating at full power).

Here, the measuring instrument should be operated in the matching bandwidth range and, thus, E field strength is measured for all broadcast control channels (BCCH).

In similar fashion, field calculations can be done for other services like CDMA and WiMAX. Once all peak E field levels are available for all operators running different services, their ratios are calculated with respect to individual field quantity (E_{Limit}) limits (depending on frequency of operation). For compliance, RMS of these ratios should be less than one.

Measurement facts. Assessment of exposure levels by measurement is advantageous in many ways. In this case, effect of all radiating sources is accessed with real parameters and real environment factors including obstacles, reflection from metallic objects and high-rise buildings. Further, little knowledge of radiating source is required.

On the other side, it is difficult to confirm and check that all sources are radiating at full load. Moreover, there is no antenna/probe covering the whole frequency band. Measurement is not possible for radiating sources that do not exist yet. Effect of the presence of staff and equipment on the field distribution has to be avoided. Some sort of post-processing is required because of different limits

as, by calculations or by software simulations. Many EM simulation software like CST Microwave Studio, High Frequency Structure Simulator (HFSS), Orthoslice, IE3D, WASP-NET, MicroWave Wizard, FEKO and EM Simulator are commercially available. These are being used for microwave analysis, design and optimisation. These simulation software make use of different numerical methods like finite-difference time-domain (FDTD), method of moments (MOM), multiple-region finite-difference time-domain (MR-FDTD), numeric EM code (NEC) and ray tracing model.

There is no one EM software application or approach that solves all problems equally well. They all have their individual strengths and weaknesses. Selection of the appropriate numerical method depends on the structure to be simulated (antenna, microstrip lines, transmission lines, waveguide structures), availability of computer resources (memory), simulation time constraints, quantity to be observed (SAR/E field/H field/power density), field zone (near/far-field) where exposure assessment is required, topology (open/closed) of the surrounding, accuracy of measurements and so on (given in the table).

FDTD method can be used to predict field values in complex scattering environments by specifying appropriate boundary conditions, or to predict SAR by specifying the dielectric properties and dimensions of the human body and appropriate boundary conditions for closed or open spaces.

For geometries having sparse regions, MR-FDTD is more suitable because the problem space is subdivided into several independent sub-

for different operating frequencies.

Assessment by software simulation. Since assessment of exposure levels by measurement is not always possible due to one or more reasons, several other methods are used for the assessment, such

regions. Fields in the sub-regions are calculated with the use of localised FDTD lattices.

Ray tracing is useful for evaluation of fields in far-field region, especially in large open areas including complex scattering environments. Ray tracing does not enable calculation of SAR. So, a combination of FDTD and ray tracing is used for taking advantages of each of these, that is, calculation of SAR and other field quantities.

MOM method is also an efficient technique to evaluate field quantities in near- and far-field regions (open/closed) emanating from thin-wire conductive structures. But it is not suitable for determining field penetration through dielectric materials. So, MOM is not used for SAR assessment.

Another approach for exposure assessment is synthetic modelling of an antenna system. In such modelling, each antenna is considered to be made of several elementary sources (say, dipole antenna). And each of these is considered as a separate radiating source. But, accuracy is somewhat lower due to negligence of coupling between elementary radiating sources.

For calculations of field quantities, point source model of the antenna system can also be used, but with some limitations. In this model, the radiating source is represented by one-point source situated in the antenna electric centre, having a radiation pattern of the considered radiating source. Boundaries of field regions have to be calculated by taking the actual size of the radiating source. This is because this model does not take into account antenna size. Hence, applicability of this model, especially in near-field calculations, is limited. If the results of calculations are to be accurate, minimum distance between the point of investigation and transmitting antenna has to fulfil requirements for the far-field region (distance = $\max(3\lambda, 2D^2/\lambda)$).

Assessment by calculations.

Assessment of exposure by EIRP calculations. Calculation of equivalent isotropically radiated power (EIPR) in the direction of maximum antenna gain is an efficient method for the assessment of exposure levels of a